

CSC111 Assembly Language Programming

Fall 2025 Midterm Exam ANS

Student Name: _____ ID: _____

Total Points	
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Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
/10	/10	/20	/10	/10	/10	/20	/10

(In all the problems, the system is assumed to be 32-bit unless specified otherwise.)

Q1 Multiple-choice questions: enter your answer keys here:

1	2	3	4	5	6	7	8	9	10
A	C	B	C	B	C	C	C	B	A

For the following multiple-choice questions, each question has exactly one correct answer key. If multiple choices are correct, choose the option “All of the above”. **Fill in the answer keys in the table above.**

(Answer keys written in the question area will not be counted.)

- Which registers are general-purpose on ARM Cortex-M?
A. R0–R12
B. R13–R15
C. CONTROL, BASEPRI, PRIMASK
D. xPSR only
ANS: A
- What are the roles of R13, R14, and R15 on ARM Cortex-M?
A. General purpose
B. Program counter, stack pointer, and link register in that order
C. Stack pointer (R13), link register (R14), program counter (R15)
D. Interrupt mask registers
ANS: C
- On ARM Cortex-M3, the borrow and carry flags relation is:
A. Carry = Borrow
B. Carry = NOT Borrow
C. Borrow always 0
D. Carry always 0
ANS: B

-
4. In two's complement, $TC(x)$ can be obtained by:
- A. Invert bits
 - B. Invert bits and subtract one
 - C. Invert bits and add one
 - D. Add one then invert bits
- ANS: C
5. In a 5-bit system, which statement is true about -16 (10000_2)?
- A. Its two's complement is 00000_2
 - B. Its two's complement is itself
 - C. It cannot be represented
 - D. It equals $+16$
- ANS: B (most negative number maps to itself)
6. What is the bit width of each register in ARM Cortex-M processors?
- A) 16 bits
 - B) 24 bits
 - C) 32 bits
 - D) 64 bits
- ANS: C) 32 bits
7. In ARM assembly instruction format, what is typically the first operand (operand1)?
- A) Source register
 - B) Immediate value
 - C) Destination register
 - D) Memory address
- ANS: C) Destination register
8. When adding two 64-bit integers split across two registers each, which instruction pair correctly handles the low and high halves?
- A. ADC for high halves, then ADDS for low halves to set carry
 - B. ADC for low halves, then ADDS for high halves
 - C. ADDS for low halves to set carry, then ADC for high halves
 - D. ADD for low halves, then ADD for high halves
- ANS: C (the order cannot be reversed)
9. In ARM subtraction, what does the carry flag C indicate when a borrow occurs in SUBS?
- A. $C = 1$ when there is a borrow
 - B. $C = 0$ when there is a borrow
 - C. C toggles regardless of borrow
 - D. C is always preserved from the previous instruction
- ANS: B (Carry equals not Borrow)
10. Which single instruction multiplies a register by 17 using the barrel shifter on the second operand?
- A. ADD r4, r4, r4, LSL #4
 - B. RSB r5, r5, r5, LSL #5
 - C. ADD r1, r0, r0, ASR #3
 - D. MUL r1, r0, #17
- ANS: A

Q2. (10 points) Assuming a 4-bit system. Show the equivalent decimal values when the data is interpreted as unsigned binary or signed binary.

Binary Value	Signed Decimal Value	Unsigned Decimal Value
1111		
1011		
0110		
1001		
0011		

ANS:

Binary Value	Signed Decimal Value	Unsigned Decimal Value
1111	-1	15
1011	-5	11
0110	6	6
1001	-7	9
0011	3	3

Q3. (20 points)

(a) (5 points) Assume an 8-bit system. Copy bits 5..3 in R4 to be the rightmost bits of R5, and set the other R5 bits to 0.

ANS:

```
MOV R0, #0x38      @ build mask 00111000
Or:
MOV R0, #7 << 3    @ build mask 00111000
Or:
MOV R0, #7          @ R0 = 00000111
LSL R0, R0, #3      @ R0 = 00111000 (now mask is 0x38)

AND R5, R4, R0      @ isolate bits 5..3. Or: AND R5, R4, #0x38
LSR R5, R5, #3      @ shift them to bits 2..0
```

Alternatively, compute $R5 = (R4 \gg 3) \& 0x7$ (shift right then isolate the lower 3 bits)

```
MOV R5, R4, LSR #3 @ R5 = R4 >> 3
Or:
LSR R5, R4, #3     @ R5 = R4 >> 3

AND R5, R5, #0x7   @ keep only lower 3 bits
```

(b) (5 points) Assume an 8-bit system. Copy the bottom four bits 3..0 in R0 to bits 6..3 in R1, and keep the other bits in R1 unchanged.

```

AND  R2, R0, #0x0F    ; take bits 3..0 from R0 into R2
LSL  R2, R2, #3       ; move into positions 6..3 in R2
AND  R1, R1, #0x87    ; clear bits 6..3 in R1 with mask 0x87 = 10000111
                        ; (preserve other bits of R1)

Or:
BIC  R1, R1, #0x78    ; clear bits 6..3 in R1 with mask 0x78 = 01111000

ORR  R1, R1, R2        ; insert new bits

```

ANS:

Or: use instruction BFI <Rd>, <Rn>, #<lsb>, #<width>

Semantics: Takes the lowest <width> bits from <Rn>. Inserts them into <Rd>, starting at bit position <lsb>. Bits outside the range [lsb, lsb+width-1] of <Rd> remain unchanged.

```

BFI R1, R0, #3, #4    ; copy 4 bits from R0 (starting at bit 0) into R1 at bit
position 3

```

(c) (10 points) Assume R0 and R1 are initialized with the values below. Write a sequence of assembly instructions that would produce the values in registers R2–R5, from R0 and R1.

Given:

R0	0xCAFEFADE
R1	0x00FF00FF

Produce:

R2	0xCAFFFAFF
R3	0xCAFFFA8F
R4	0xFFFFFCA
R5	0xDEFF00FF

ANS:

```

ORR R2, R0, R1      @ R2 = 0xCAFEFADE | 0x00FF00FF = 0xCAFFFAFF
BIC R3, R2, #0x70    @ R3 = 0xCAFFFAFF & not 01110000 = 0xCAFFFA8F
ASR R4, R0, #24      @ R4 = 0xCAFEFADE >> 24 = 0xFFFFFCA (sign-extended)

```

For R5, option 1:

```

LSL R5, R0, #24      @ R5 = 0xCAFEFADE << 24 = 0xDE000000
ORR R5, R5, R1      @ R5 = 0xDE000000 | 0x00FF00FF = 0xDEFF00FF

```

For R5, option 2:

```

ORR R5, R1, R0, LSL #24

```

For R5, option 3:

```

MOV R5, R1           @ R5 = 0x00FF00FF
BFI R5, R0, #24, #8  @ Bit Field Insert - insert the lowest 8 bits of R0
bits [7:0] (0xDE) into bits [31:24] of R5

```

Note that this instruction is invalid due to constraints on the immediate value (c.f., last few slides of Ch4 ARM Arithmetic Logic):

```

AND R3, R2, #0xFFFFFFFF8F

```

Q4. (10 points) Assume a 4-bit system. For each of the following operations, compute the result and NZCV flags, based on the first row that has been given to you.

Operation	Result in binary	Equivalent unsigned arithmetic in decimal	Equivalent signed arithmetic in decimal	NZCV
1001 + 0010	1011	9 + 2 = 11	-7 + 2 = -5	1000
1101 + 1100				
1101 - 1100				
1100 + 1010				
0100 - 0110				
0100 + 0010				

ANS: (Unsigned 4-bit range: 0 to 15, Signed 4-bit range: -8 to +7. In case of overflow, please write the actual result after truncation to 4 bits. In this exam, you are also given credit for the ground truth result that exceeds the range.)

Operation	Result in binary	Equivalent unsigned arithmetic in decimal	Equivalent signed arithmetic in decimal	NZCV
1001 + 0010	1011	9 + 2 = 11	-7 + 2 = -5	1000
1101 + 1100	1001	13 + 12 = 9 (or 25)	-3 + (-4) = -7	1010
1101 - 1100	0001	13 - 12 = 1	-3 - (-4) = 1	0010
1100 + 1010	0110	12 + 10 = 6	-4 + (-6) = 6 (or -10)	0011
0100 - 0110	1110	4 - 6 = 14 (or -2)	4 - 6 = -2	1000
0100 + 0010	0110	4 + 2 = 6	4 + 2 = 6	0000

Key Explanations:

1101 + 1100: Result wraps to 1001. Carry out occurs (C=1), but no signed overflow since -3 + (-4) = -7 is valid in 4-bit signed range.

1101 - 1100: Straightforward subtraction giving 1. No borrow needed (C=1).

1100 + 1010: Adding two negative numbers (-4 + -6 = -10) produces a positive result (+6) because -10 is outside the 4-bit signed range [-8, 7]. This sets overflow V=1.

0100 - 0110: Subtracting larger from smaller requires a borrow (C=0). Result is 1110 = -2 in signed, 14 in unsigned.

0100 + 0010: Clean addition with no flags set - result is positive, non-zero, no carry, no overflow.

Q5. (10 points) For each of the following instructions (executed individually, not sequentially), compute the result and NZCV flags. Assume initially R0=0xFFFFFFFF, R1=0x00000001, NZCV=0000.

Instruction	Result in dest. register R0	NZCV
MOVS R0, #0		
ANDS R0, R0, #0		
ORRS R0, R0, R1		

ANDS R0, R0, R1		
ADDS R0, R0, R1		
SUBS R0, R0, R1		
ADDS R0, R0, R1, LSL #31		s
BICS R0, R0, R1, LSL #31		
EORS R0, R0, R1, LSL #31		
LSLS R0, R0, #31		

ANS: (Assume initially R0=0xFFFFFFFF, R1=0x00000001.)

Instruction	Result in R0	NZCV	Explanation
MOVS R0, #0	0x00000000	0100	Result is zero (Z=1), bit 31=0 (N=0), MOVS doesn't affect C/V
ANDS R0, R0, #0	0x00000000	0100	Any value AND 0 = 0, so Z=1, N=0, logical ops don't affect C/V
ORRS R0, R0, R1	0xFFFFFFFF	1000	0xFFFFFFFF 0x00000001 = 0xFFFFFFFF, bit 31=1 (N=1), non-zero (Z=0)
ANDS R0, R0, R1	0x00000001	0000	0xFFFFFFFF & 0x00000001 = 0x00000001, bit 31=0 (N=0), non-zero (Z=0)
ADDS R0, R0, R1	0x00000000	0110	0xFFFFFFFF + 0x00000001 = 0 with carry out (C=1), Z=1, no overflow
SUBS R0, R0, R1	0xFFFFFFFFFE	1010	0xFFFFFFFF - 0x00000001 = 0xFFFFFFFFFE, N=1, no borrow (C=1)
ADDS R0, R0, R1, LSL #31	0x7FFFFFFF	0011	R1<<31=0x80000000; 0xFFFFFFFF+0x80000000=0x7FFFFFFF; Overflow! (V=1), Carry (C=1)
BICS R0, R0, R1, LSL #31	0x7FFFFFFF	0000	R0 & ~(0x80000000) = 0x7FFFFFFF, clears bit 31, N=0, Z=0
EORS R0, R0, R1, LSL #31	0x7FFFFFFF	0000	0xFFFFFFFF ^ 0x80000000 = 0x7FFFFFFF, flips bit 31, N=0, Z=0
LSLS R0, R0, #31	0x80000000	1010	0xFFFFFFFF<<31=0x80000000, N=1, last bit out was 1 (C=1)

Key Explanations:

- 1) ADDS R0, R0, R1, LSL #31 causes **signed overflow** (V=1) when adding two negative numbers that produce a positive result
- 2) BICS R0, R0, R1, LSL #31 and EORS R0, R0, R1, LSL #31 both produce 0x7FFFFFFF by clearing/flipping bit 31
- 3) LSLS R0, R0, #31 the carry flag C receives the last bit shifted out, which for a shift of 31 is the original bit 1, which is 1.

Q6. (10 points)

(a) Suppose r0 = 0x00008000, and the following memory layout:

Address	Data
0x00008007	0x79

0x00008006	0xCD
0x00008005	0xA3
0x00008004	0xFD
0x00008003	0x0D
0x00008002	0xEB
0x00008001	0x2C
0x00008000	0x1A

(a1) (3 points) ARM processors can be configured as big-endian or little-endian. What is the value of r1 after running LDR r1, [r0]?

a. If little-endian, r1 = _____

b. If big-endian, r1 = _____

ANS:

a. If little-endian, r1 = **0x0DEB2C1A**

b. If big-endian, r1 = **0x1A2CEB0D**

(a2) (3 points) Suppose the system is little-endian. What are the values of r1 and r0? Each instruction is executed individually, not sequentially.

a. LDR r1, [r0, #4]

b. LDR r1, [r0], #4

c. LDR r1, [r0, #4]!

After LDR r1, [r0, #4]

r0 = _____

r1 = _____

After LDR r1, [r0], #4

r0 = _____

r1 = _____

After LDR r1, [r0, #4]!

r0 = _____

r1 = _____

ANS:

After LDR r1, [r0, #4]

r0 = **0x00008000**

r1 = **0x79CDA3FD**

After LDR r1, [r0], #4

r0 = **0x00008004**

$r1 = 0x0DEB2C1A$
 After `LDR r1,[r0,#4]!`
 $r0 = 0x00008004$
 $r1 = 0x79CDA3FD$

Explanations:

- 1) `LDR r1, [r0, #4]` — loads from `r0+4` but **does not** change `r0`.
- 2) `LDR r1, [r0], #4` — post-index; load from `r0`, then `r0 += 4`.
- 3) `LDR r1, [r0, #4]!` — pre-index; `r0 += 4` then load from new `r0`

(b) (4 points) Assume little-endian memory ordering. Suppose `r0 = 0x2000,0000` and `r1 = 0x12345678`. All bytes in memory are initialized to `0x00`. Fill the following table after the assembly program has finished execution.

<code>STR r1, [r0], #4</code>
<code>STR r1, [r0, #4]!</code>
<code>STR r1, [r0]</code>

ANS:

Memory Address	Initial Memory Content	Final Memory Content
0x2000,0013	00	00
0x2000,0012	00	00
0x2000,0011	00	00
0x2000,0010	00	00
0x2000,000F	00	00
0x2000,000E	00	00
0x2000,000D	00	00
0x2000,000C	00	00
0x2000,000B	00	12
0x2000,000A	00	34
0x2000,0009	00	56
0x2000,0008	00	78
0x2000,0007	00	00
0x2000,0006	00	00
0x2000,0005	00	00
0x2000,0004	00	00
0x2000,0003	00	12
0x2000,0002	00	34
0x2000,0001	00	56
0x2000,0000	00	78

Q7. (20 points) Show all updates to registers as the assembly code shown below runs, assuming **big-endian** ordering. (Memory addresses increase from left to right in the table.) Fill in the tables of final register values and memory contents. Show the NZCV flags after execution, assuming NZCV=0000 initially.

Assembly program

```

LDR R1, =0x10000010
LDR R2, [R1]
LDR R3, [R1, #4]!
ASR R4, R2, #12
MOVW R5, #14
LSL R6, R5, #9
BIC R7, R2, R6
AND R8, R7, R6
STR R7, [R1, #8]

```

Initial memory contents:

0x10000010	FF	EF	CD	AB	00	00	CD	AB	AA	BB	CC	DD	EE	FF	11	22
------------	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

Final register values:

R1		R2		R3		R4	
R5		R6		R7		R8	

Final memory contents:

0x10000010																
------------	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Final NZCV = _____

ANS:

Registers

R1	0x10000014	R2	0xFFEFCDA B	R3	0x0000CDAB	R4	0xFFFFFEFC
R5	0x0000000E	R6	0x00001C00	R7	0xFFEFC1AB	R8	0x00000000

Final memory contents:

0x10000010	FF	EF	CD	AB	00	00	CD	AB	AA	BB	CC	DD	FF	EF	C1	AB
------------	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

Final NZCV = 1010

Explanations (not required for exam):

1. LDR R1, =0x10000010 → R1 = 0x10000010.
2. LDR R2, [R1] → load bytes FF EF CD AB → R2 = 0xFFEFCDA B.
3. LDR R3, [R1, #4]! → **pre-index**: it increments R1 to 0x10000014 before the load. Load 00 00 CD AB → R3 = 0x0000CDAB.
4. ASR R4, R2, #12 → arithmetic shift right 12 of 0xFFEFCDA B → R4 = 0xFFFFFEFC. **NZCV changes here**: N=1, Z=0, C=bit11(R2)=1, V unchanged → NZCV = 1010.
5. MOVW R5, #14 → R5 = 0x0000000E.
6. LSL R6, R5, #9 → R6 = 0x00001C00.
7. BIC R7, R2, R6 → R7 = R2 & ~R6 = 0xFFEFC1AB.
8. AND R8, R7, R6 → R8 = 0x00000000.
9. STR R7, [R1, #8] → stores at address R1 + 8 = 0x10000014 + 8 = 0x1000001C. Write 0xFFEFC1AB (big-endian bytes FF EF C1 AB) to 0x1000001C..0x1000001F.

Final memory (address order from 0x10000010):

- 0x10000010..0x10000013 = FF EF CD AB
- 0x10000014..0x10000017 = 00 00 CD AB

- 0x100000018..0x10000001B = AA BB CC DD (unchanged)
- 0x10000001C..0x10000001F = FF EF C1 AB (stored value)

Q8. (10 points) Consider an array[] of 25 integers (each integer is 4 bytes). Assume register R2 holds the base memory address of array, i.e., array[0]. A compiler associates variables x and y with registers R0 and R1, respectively. Translate this C program into ARM assembly language based on the provided comments.. (The original problem did not specify if x is unsigned or signed integer, so both LSR and ASR are graded as correct for computing x/4.)

C program	Assembly program
uint32_t x = array[5] + y;	_____ ; load array[5] (with 5*4 = 20 bytes offset) into _____ ; temporary register R3
array[6] = x * 4;	_____ ; x = array[5] + y
array[7] = x / 4;	_____ ; R3 = x * 4
array[8] = x - 10;	_____ ; store into array[6]
array[9] = x * (x - 1);	_____ ; R3 = x / 4
	_____ ; store into array[7]
	_____ ; R3 = x - 10
	_____ ; store into array[8]
	_____ ; R4 = x - 1
	_____ ; R3 = x * (x - 1)
	_____ ; store into array[9]

ANS:

C program	Assembly program
uint32_t x = array[5] + y; array[6] = x * 4; array[7] = x / 4; array[8] = x - 10; array[9] = x * (x - 1);	LDR R3, [R2, #20] ; load array[5] (with 5*4 = 20 bytes offset) into a temp register ADD R0, R3, R1 ; x = array[5] + y LSL R3, R0, #2 ; R3 = x * 4. Or: MOV R1, #4 MUL R3, R0, R1 ; Note that MUL R3, R0, #4 is not valid syntax STR R3, [R2, #24] ; store into array[6] LSR R3, R0, #2 ; R3 = x / 4. Note that UDIV R3, R0, #2 is not valid syntax STR R3, [R2, #28] ; store into array[7] SUB R3, R0, #10 ; R3 = x - 10 STR R3, [R2, #32] ; store into array[8] SUB R4, R0, #1 ; R4 = x - 1 MUL R3, R0, R4 ; R3 = x * (x - 1) STR R3, [R2, #36] ; store into array[9]

Or:

C program	Assembly program
uint32_t x = array[5] + y; array[6] = x * 4; array[7] = x / 4; array[8] = x - 10; array[9] = x * (x - 1);	LDR R3, [R2], #20 ; load array[5] (with 5*4 = 20 bytes offset) into a temp register ADD R0, R3, R1 ; x = array[5] + y LSL R3, R0, #2 ; R3 = x * 4. STR R3, [R2], #4 ; store into array[6] LSR R3, R0, #2 ; R3 = x / 4. Note that UDIV R3, R0, #2 is not valid syntax STR R3, [R2], #4 ; store into array[7] SUB R3, R0, #10 ; R3 = x - 10 STR R3, [R2], #4 ; store into array[8] SUB R4, R0, #1 ; R4 = x - 1 MUL R3, R0, R4 ; R3 = x * (x - 1) STR R3, [R2], #4 ; store into array[9]

Appendix; Conversion table. (You may tear off and discard this page.)

Decimal	Binary	Hex
0	0000	0x0
1	0001	0x1
2	0010	0x2
3	0011	0x3
4	0100	0x4
5	0101	0x5
6	0110	0x6
7	0111	0x7
8	1000	0x8
9	1001	0x9
10	1010	0xA
11	1011	0xB
12	1100	0xC
13	1101	0xD
14	1110	0xE
15	1111	0xF